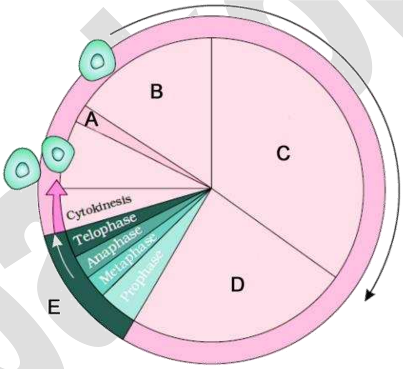
10. CELL CYCLE AND CELL DIVISION

1. The growth and reproduction of all organisms depend on
   a. The division of cells and tissues
   b. The division and enlargement of cells
   c. The division & enlargement of cells and tissues
   d. The enlargement of cells
2. Duration of cell cycle in a typical eukaryotic cell such as human cell is about
   a. 24 minutes b. 90 minutes
   c. 24 hours d. 90 hrs
3. The two basic phases of cell cycle are
   a. Mitosis & meiosis
   b. Mitosis & Interkinesis
   c. Interphase & M phase
   d. G_1 and G_2 phase
4. The phase between two successive M phases is called
   a. G_1 phase b. Antephase
   c. Karyokinesis d. Interphase
5. Three phases involved in interphase are
   a. G_1 phase, S phase, G_2 phase
   b. G_1 phase, G_2 phase, M phase
   c. G_1 phase, karyokinesis, cytokinesis
   d. G_1 phase, S phase, M phase
6. Select the false statement
   a. Interphase lasts more than 95% of the duration of cell cycle
   b. Interphase includes cell division & DNA synthesis
   c. S phase is the second phase of interphase and is the longest phase
   d. In both G_1 & G_2 phases, synthesis of RNA and proteins occurs
7. Which is correctly labelled?



   a. A= G_1 phase B= S phase C= G_0 phase D= G_2 phase E= M phase
   b. A= G_0 phase B= S phase C= G_1 phase D= G_2 phase E= M phase
   c. A= M phase B= G_0 phase C= G_1 phase D= S phase E= G_2 phase
   d. A= G_0 phase B= G_1 phase C= S phase D= G_2 phase E= M phase
8. Which of the following represents actual cell division?
   a. G_1 phase b. S phase
   c. G_2 phase d. M phase
9. The cells that do not divide further exit G_1 phase and enter an inactive stage called
   a. Quiescent (G_0) stage b. G_2 phase
   c. Synthetic (S) phase d. Early prophase
10. The cells that do not show division are
    a. Epithelial cells, heart cells and megakaryocytes
    b. Neurons, skeletal and cardiac muscle cells
    c. Neurons, bone marrow cells and muscle cells
    d. Hemocytoblasts, heart cells and epithelial cells
11. Given below are some statement regarding mitosis Select the wrong one.
    a. It is the cell division occurring in somatic cells
    b. It is also called as equational division as the number of chromosomes in the parent and progeny cells is same.
    c. It occurs only in diploid cells.
    d. It helps to retain the same chromosome number in all somatic cells.
12. The correct sequence of stages in karyokinesis is
    a. Metaphase, Prophase, Telophase & Anaphase
    b. Prophase, Metaphase, Anaphase & Telophase
    c. Telophase, Prophase, Anaphase & Metaphase
    d. Anaphase, Prophase, Telophase & Metaphase
13. The longest and shortest phases in mitosis is
    a. Prophase and Anaphase respectively
    b. Metaphase and Telophase respectively
    c. Anaphase and Prophase respectively
    d. Telophase and Metaphase respectively
14. In animal cells, during S phase, DNA replication begins in the ------ and the centriole duplication in the ------
    a. Nucleus, cytoplasm b. Nucleus, nucleus
    c. Cytoplasm, nucleus d. Cytoplasm, cytoplasm
15. Match the following

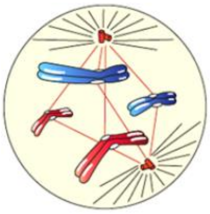
A	B
A. Prophase	1. Chromosomes at equator
B. Metaphase	2. Uncoiling of chromosomes
C. Anaphase	3. Condensation of chromosomes
D. Telophase	4. Chromatids move to opposite poles

	A	B	C	D
a.	2	3	4	1
b.	4	3	2	1
c.	3	4	1	2
d.	3	1	4	2

16. Which of the following does not occur during late prophase?
    a. Each chromosome splits into two chromatids attached together at the centromere
    b. The nuclear envelope, nucleolus, Golgi complexes & endoplasmic reticulum disappear
    c. Chromosomes form two sister chromatids
    d. Condensation of chromosomes continues
17. In animal cells, during late prophase, the centrioles
    a. Move to opposite poles and radiate out astral rays

b. Move to equator and radiate out astral rays
c. Move to equator and radiate out spindle fibres
d. Move to opposite poles and start uncoiling of chromosomes

18. Regarding prophase
    1. Prophase is the first stage of mitosis
    2. Prophase follows S and G2 phases of interphase
    3. Prophase is marked by the initiation of condensation of chromosomal material.
    4. The centriole which had undergone duplication during S phase of interphase, now begins to move towards opposite poles of the cell
    a. All are correct statements
    b. Only 1,3 are correct statements
    c. Only 2,4 are correct statements
    d. Only 1,2,3 are correct statements
19. The spindle fibres disappear during
    a. Prophase   b. Metaphase
    c. Anaphase   d. Telophase
20. The figure given below represents



    a. Transition to late Prophase
    b. Transition to Metaphase
    c. Transition to Anaphase
    d. Transition to Telophase
21. Which of the following options gives the correct sequences of events during mitosis?
    a. Condensation → nuclear membrane disassembly → crossing over → segregation → telophase
    b. Condensation → nuclear membrane disassembly → arrangement at equator → centromere division → segregation → telophase
    c. Condensation → crossing over → nuclear membrane disassembly → segregation → telophase
    d. Condensation → arrangement at equator → centromere division → segregation → telophase
22. Which of the following is not the significance of mitosis?
    a. It produces diploid daughter cells with identical genome
    b. It helps to retain the same chromosome number in all somatic cells
    c. It restores the nucleo-cytoplasmic ratio that disturbed due to cell growth
    d. It causes genetic variation due to crossing over in the population of organisms
23. Cytokinesis in plant cell is different from animal cell due to the presence of
    a. Large vacuole   b. Cell wall
    c. Chloroplasts   d. All of these
24. Which is not occurring in cytokinesis of plants?
    a. The vesicles formed from Golgi bodies accumulate at the equator
    b. Cell plate is formed
    c. A cleavage furrow is appeared in the plasma membrane
    d. The cell plate becomes the middle lamella
25. Meiosis occurs during
    a. Gametogenesis   b. Body growth
    c. Tissue repair   d. Crossing over
26. Which of the following is not the key feature of meiosis?
    a. It involves pairing of homologous chromosomes and recombination between them.
    b. It involves two cycles of DNA replication and cell division.
    c. Meiosis I begins after replication of parental chromosomes to form identical sister chromatids at the S phase
    d. Four haploid cells are formed at the end of meiosis II
27. Choose the correct sequence of events of Prophase I?
    a. Zygotene, Leptotene, Diplotene, Pachytene & Diakinesis
    b. Pachytene, Diplotene, Leptotene, Zygotene & Diakinesis
    c. Diplotene, Leptotene, Zygotene, Pachytene & Diakinesis
    d. Leptotene, Zygotene, Pachytene, Diplotene & Diakinesis
28. Match the following

A	B
A. Leptonema	1. Fully condensed chromosomes
B. Zygonema	2. Crossing over
C. Pachynema	3. Synapsis
D. Diplonema	4. Long slender chromosomes
E. Diakinesis	5. Chiasmata

	A	B	C	D	E
a.	2	3	5	1	4
b.	4	3	2	5	1
c.	5	4	1	2	3
d.	3	1	4	5	2

29. During zygotene, similar chromosomes start pairing together. It is called
    a. Crossing over   b. Recombination
    c. Synapsis   d. Diakinesis
30. Each pair of homologous chromosomes is called a
    a. Bivalent   b. Tetrad
    c. Chiasmata   d. Diad
31. Crossing over is
    a. The pairing of homologous chromosomes with the help of a complex structure called synaptonemal complex.
    b. The exchange of genetic material between sister chromatids of a chromosomes in presence of *recombinase.*
    c. The pairing of non-homologous chromosomes with the help of a complex structure called synaptonemal complex.

d. The exchange of genetic material between non-sister chromatids of two homologous chromosomes in presence of *recombinase.*

32. Kinetochore is located in
    a. Centriole b. Centromere
    c. Chromatids d. Lysosome
33. Recombination is completed by the end of
    a. Zygotene b. Pachytene
    c. Diplotene d. Diakinesis
34. This short stage between the two meiotic divisions is called
    a. Interphase b. Antephase
    c. Interkinesis d. Cytokinesis
35. "Thin threaded" stage refers to
    a. Leptotene b. Zygotene
    c. Pachytene d. Diplotene
36. Number of chromatids at metaphase is
    a. Two each in mitosis and meiosis
    b. Two in mitosis and one in meiosis
    c. Two in mitosis and four in meiosis
    d. One in mitosis and two in meiosis
37. Total number of reproductive cells required for the formation of 128 gametes in man is
    a. 32 b. 64 c. 16 d. 128
38. Match the following

	Column A		Column B
1	Mitosis in plants	a	Inner cheek cells
2	Meiosis in plants	b	Grasshopper testis
3	Mitosis in animals	c	In plants
4	Meiosis in animals	d	Onion root tip
5	Cell plate formation	e	In animals
6	Cleavage furrow	f	Inhibits spindle fibres leads to polyploidy
7	Colchicine	g	Young flower buds in tradescantia

a. 1-a, 2-b, 3-c, 4-d, 5-e, 6-f, 7-g
b. 1-d, 2-g, 3-c, 4-e, 5-b, 6-a, 7-f
c. 1-d, 2-g, 3-a, 4-b, 5-c, 6-e, 7-f
d. 1-d, 2-g, 3-a, 4-e, 5-c, 6-f, 7-b

>> Answer Key: Click Here

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